9.18.

$$E_{n_x n_y n_z} = \frac{\hbar^2 \pi^2}{2ma^2}\left(n_x^2 + n_y^2 + n_z^2\right)$$

Ground state:

$n_x = 1;\ n_y = 1;\ n_z = 1 \qquad E = \dfrac{3\hbar^2\pi^2}{2ma^2} \qquad degeneracy = 1$

First excited state:

$n_x = 2;\ n_y = 1;\ n_z = 1 \qquad E = \dfrac{3\hbar^2\pi^2}{ma^2} \qquad degeneracy = 3$

$n_x = 1;\ n_y = 2;\ n_z = 1$

$n_x = 1;\ n_y = 1;\ n_z = 2$

Second excited state:

$n_x = 2;\ n_y = 2;\ n_z = 1 \qquad E = \dfrac{9\hbar^2\pi^2}{2ma^2} \qquad degeneracy = 3$

$n_x = 2;\ n_y = 1;\ n_z = 2$

$n_x = 1;\ n_y = 2;\ n_z = 2$

9.19.

a)

$$\Delta x \Delta p = \frac{\hbar}{2}\sqrt{\frac{n^2\pi^2}{3} - 2}$$

n=1

$$\Delta x \Delta p = \frac{\hbar}{2}\sqrt{\frac{\pi^2}{3} - 2} \approx 0.568\hbar$$

n=2

$$\Delta x \Delta p = \frac{\hbar}{2}\sqrt{\frac{4\pi^2}{3} - 2} \approx 1.67\hbar$$

n=3

$$\Delta x \Delta p = \frac{\hbar}{2}\sqrt{\frac{9\pi^2}{3} - 2} \approx 2.63\hbar$$

b)

$$\Delta x = \frac{a}{2\pi n}\sqrt{\frac{n^2\pi^2}{3} - 2}$$

n=1
a=0.2 nm

$$\Delta x = \frac{a}{2\pi}\sqrt{\frac{\pi^2}{3} - 2} \approx 0.18\, a \approx 0.036\, nm$$

a=2 nm

$$\Delta x = \frac{a}{2\pi}\sqrt{\frac{\pi^2}{3} - 2} \approx 0.18\, a \approx 0.36\, nm$$

9.20.

$$v = 1; \qquad \Psi_1(x) = \left(\frac{4\alpha^3}{\pi}\right)^{1/4} x e^{-\frac{\alpha x^2}{2}}$$

$$\langle x \rangle = 0$$

$$\Delta x = \sqrt{\sigma_x^2} = \sqrt{\langle x^2 \rangle - \langle x \rangle^2} = \sqrt{\langle x^2 \rangle}$$

$$\langle x^2 \rangle = \int_{-\infty}^{+\infty} \Psi_1^*(x) x^2 \Psi_1(x)\, dx = \left(\frac{4\alpha^3}{\pi}\right)^{1/2} \int_{-\infty}^{+\infty} x^4 e^{-\alpha x^2}\, dx = \left(\frac{4\alpha^3}{\pi}\right)^{1/2} \left(\frac{9\pi}{16\alpha^5}\right)^{1/2} = \frac{3}{2\alpha}$$

$$\Delta x = \sqrt{\frac{3}{2\alpha}}$$

9.21.

$$\upsilon = 1; \qquad \Psi_1(x) = \left(\frac{4\alpha^3}{\pi}\right)^{1/4} x e^{-\frac{\alpha x^2}{2}}$$

$$\langle p \rangle = 0$$

$$\Delta p = \sqrt{\sigma_p^2} = \sqrt{\langle p^2 \rangle - \langle p \rangle^2} = \sqrt{\langle p^2 \rangle}$$

$$\langle p^2 \rangle = \int_{-\infty}^{+\infty} \Psi_1^*(x)\hat{p}\hat{p}\Psi_1(x)\, dx = \left(\frac{4\alpha^3}{\pi}\right)^{1/2} \int_{-\infty}^{+\infty} x e^{-\frac{\alpha x^2}{2}} (-1)\hbar^2 \frac{d^2}{dx^2} x e^{-\frac{\alpha x^2}{2}}\, dx =$$

$$= \left(\frac{4\alpha^3}{\pi}\right)^{1/2} \hbar^2 \int_{-\infty}^{+\infty} x e^{-\alpha x^2} (3\alpha x - \alpha^2 x^3)\, dx = \left(\frac{4\alpha^3\hbar^4}{\pi}\right)^{1/2} \int_{-\infty}^{+\infty} (3\alpha x^2 - \alpha^2 x^4) e^{-\alpha x^2}\, dx =$$

$$= \left(\frac{4\alpha^3\hbar^4}{\pi}\right)^{1/2} 3\alpha \frac{1}{2\alpha}\left(\frac{\pi}{\alpha}\right)^{1/2} - \left(\frac{4\alpha^3\hbar^4}{\pi}\right)^{1/2} \alpha^2 \frac{3}{4\alpha^2}\left(\frac{\pi}{\alpha}\right)^{1/2} = \frac{3\hbar^2\alpha}{2}$$

$$\Delta p = \hbar\sqrt{\frac{3\alpha}{2}}$$

9.22.

$$\Delta x \Delta p = \frac{3}{2}\hbar > \frac{1}{2}\hbar$$

9.23.

$^{12}C^{16}O$

$$\tilde{\nu} = 2169.814\ \mathrm{cm}^{-1}$$

$$k = \omega^2\mu = (2\pi\nu)^2\mu = (2\pi c\tilde{\nu})^2\mu$$

$$\mu = \frac{m_C m_O}{m_C + m_O} \approx \frac{12 \cdot 16}{12 + 16} 1.66 \cdot 10^{-27}\mathrm{kg} \approx 1.14 \cdot 10^{-26}\mathrm{kg}$$

$$k = (2 \cdot 3.14 \cdot 3 \cdot 10^{10}\mathrm{cm} \cdot \mathrm{s}^{-1} \cdot 2169.814\ \mathrm{cm}^{-1})^2 1.14 \cdot 10^{-26}\mathrm{kg} \approx 1.9 \cdot 10^3\mathrm{Nm}^{-1}$$

9.24.

$$\tilde{\nu} \sim \sqrt{\frac{1}{\mu}}$$

$^{12}C^{16}O \qquad \mu_1 = \frac{12 \cdot 16}{12+16} \approx 6.857 \text{ a. m. u.}$

$^{12}C^{18}O \qquad \mu_2 = \frac{12 \cdot 18}{12+18} \approx 7.2 \text{ a. m. u.}$

$$\tilde{\nu}_2 = \tilde{\nu}_1 \sqrt{\frac{\mu_1}{\mu_2}} \approx 2169.8 \sqrt{\frac{6.857}{7.2}} \approx 2117.5 \text{ cm}^{-1}$$

9.25.

$$\Delta x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2} = \sqrt{\frac{1}{2\alpha}} = \sqrt{\frac{\hbar}{2\mu\omega}} = \sqrt{\frac{\hbar}{4\mu\pi c\tilde{\nu}}} \approx$$

$$\approx \sqrt{\frac{1.05 \cdot 10^{-34}}{4 \cdot 1.14 \cdot 10^{-26} \cdot 3.14 \cdot 3 \cdot 10^{10} \cdot 2169.8}} \approx 3.36 \text{ pm}$$

Bond length: $l = 112.832$ pm

$\Delta x \approx 2.98\% \cdot l$

9.48.

a)

$\Psi = xe^{-\alpha x^2}$

$$\frac{d^2\Psi}{dx^2} - 4\alpha^2 x^2 \Psi = \frac{d}{dx}\left(e^{-\alpha x^2} - 2\alpha x^2 e^{-\alpha x^2}\right) - 4\alpha^2 x^3 e^{-\alpha x^2} =$$

$$= -2\alpha x e^{-\alpha x^2} - 4\alpha x e^{-\alpha x^2} + 4\alpha^2 x^3 e^{-\alpha x^2} - 4\alpha^2 x^3 e^{-\alpha x^2} = -6\alpha x e^{-\alpha x^2} = -6\alpha\Psi$$

Eigenvalue: -6α

b)

$\Psi = Ke^{r/k}$

$$\frac{d\Psi}{dr} = \frac{K}{k} e^{r/k} = \frac{1}{k}\Psi$$

Eigenvalue: $\frac{1}{k}$

9.50.

$$E_n = \frac{\hbar^2\pi^2}{2ma^2}n^2 = \frac{h^2}{8ma^2}n^2 \approx \frac{(6.63 \cdot 10^{-34}\,\mathrm{J\,s})^2}{8 \cdot 9.1 \cdot 10^{-31}\mathrm{kg}(0.5 \cdot 10^{-9}\mathrm{m})^2}n^2 \approx n^2 2.4 \cdot 10^{-19}\mathrm{J}$$

Energy per mole:

$$\varepsilon_n = E_n \cdot N_A \approx n^2 2.4 \cdot 10^{-19} \cdot 6 \cdot 10^{23}\mathrm{Jmol^{-1}} \approx n^2 144\ \mathrm{kJ\ mol^{-1}}$$

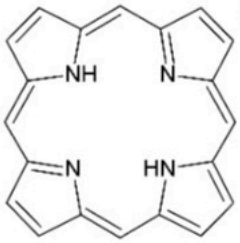
$$\varepsilon_1 = 144\ \mathrm{kJ\ mol^{-1}}$$

$$\varepsilon_2 = 576\ \mathrm{kJ\ mol^{-1}}$$

$$\varepsilon_3 = 1296\ \mathrm{kJ\ mol^{-1}}$$

Bonus Problem 1. (10 pts)

Many proteins contain porphyrin molecules. The general structure of a porphyrin molecule is shown below. This molecule is planar so we can approximate the π electrons as being confined inside a 2D square. The porphyrin molecule has 18 π electrons. Each energy level can contain up to 2 electrons.



a) (4 pts) Express the general formula for energy levels of a particle in a square of side *a*. How many energy levels are occupied by the π electrons in porphyrin? What are their quantum numbers and degeneracies?

b) (2 pts) Estimate the side length of the porphyrin molecule (C-C bond is 154 pm, C=C bond is 135 pm and C atom radius is 77 pm).

c) (4 pts) Calculate the lowest possible absorption energy of the porphyrin molecule (transition from HOMO to LUMO). Compare with the experimental value 17,000cm^{-1}

Solution:

a)

$$E_{n_x n_y} = \frac{\hbar^2 \pi^2}{2ma^2}\left(n_x^2 + n_y^2\right)$$

n_x=1 n_y=1

ground state, non-degenerate, 2 electrons

n_x=1 n_y=2; n_x=2 n_y=1

first excited state, doubly degenerate, 4 electrons

n_x=2 n_y=2

second excited state, non-degenerate, 2 electrons

n_x=1 n_y=3; n_x=3 n_y=1

third excited state, doubly degenerate, 4 electrons

n_x=2 n_y=3; n_x=3 n_y=2

fourth excited state, doubly degenerate, 4 electrons

n_x=1 n_y=4; n_x=4 n_y=1

fifth excited state, doubly degenerate, 2 electrons

b)

a=2*154+2*135+2*77=0.73 nm (if calculated more accurately it should be close to 0.6 nm)

c)

HOMO - n_x=1 n_y=4

LUMO - n_x=3 n_y=3

$$\Delta E = E_{LUMO} - E_{HOMO} = \frac{\hbar^2 \pi^2}{2ma^2}\left(3^2 + 3^2 - 1^2 - 4^2\right) = \frac{\hbar^2 \pi^2}{2ma^2} \approx$$

$$\approx \frac{(1.055 \cdot 10^{-34}\,\mathrm{J\,s})^2 \pi^2}{2 \cdot 9.1 \cdot 10^{-31}\,\mathrm{kg} \cdot (0.73 \cdot 10^{-9}\,\mathrm{m})^2} \approx 1.13 \cdot 10^{-19}\,\mathrm{J} \approx 5.7 \cdot 10^3\,\mathrm{cm}^{-1}$$

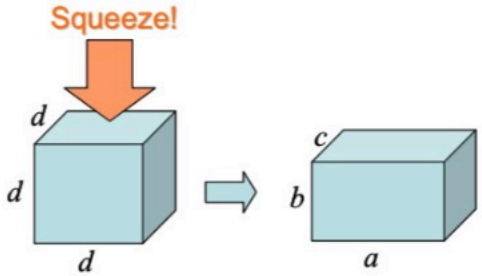
Bonus Problem 2. (10 pts)

In this problem, we will model a quantum dot (e.g., a CdSe nanoparticle) as a particle (electron) in a cubic box with side d. Consider what happens if we apply some force to one of the facets so that the cube gets distorted, but its volume remains the same ($V=d^3$):

$$a = d(1+\delta_a), \quad b = d(1-\delta_b), \quad c = d$$

$$abc = V = d^3 = const$$

(Hint: express δ_b through δ_a and use Taylor series expansion considering that $\delta_a << 1$)



a) (2 pts) What happens to the ground state energy?
b) (2 pts) Calculate the energies of the first 3 states of the distorted box.
c) (2 pts) What happens to the degeneracy of the 1st excited state of the original cubic box when it is distorted?
d) (2 pts) Calculate the energies of the states resulting from the 1st excited state of the unperturbed system (cubic box).
e) (2 pts) Discuss what happens with the absorption spectrum of this system – in particular the line shape of the first transition of the cubic box – when the distortion is applied.

This problem illustrates one of the general principles of quantum mechanics: ***degeneracy*** of states results from ***symmetry***. When the symmetry is ***lowered***, the degeneracy is ***lifted***.

Solution:

$$V = abc = d^3 = d(1+\delta_a)d(1-\delta_b)d \;\; => \;\; 1-\delta_b = \frac{1}{1+\delta_a}$$

$$E_{n_x n_y n_z} = \frac{\hbar^2\pi^2}{2m}\left(\frac{n_x^2}{a^2} + \frac{n_y^2}{b^2} + \frac{n_z^2}{c^2}\right)$$

a)
For cubic box:

$$E_{111} = \frac{3\hbar^2\pi^2}{2md^2}$$

For squeezed box:

Taylor series expansion for small δ_a

$$E_{111} = \frac{\hbar^2\pi^2}{2md^2}\left(\frac{1}{(1+\delta_a)^2} + (1+\delta_a)^2 + 1\right) \approx$$

$$\approx \frac{\hbar^2\pi^2}{2md^2}(1 - 2\delta_a + 3\delta_a^2 + 1 + 2\delta_a + \delta_a^2 + 1) \approx \frac{\hbar^2\pi^2}{2md^2}(3 + 4\delta_a^2)$$

Ground state energy will increase

b)

$$E_{112} = \frac{\hbar^2\pi^2}{2md^2}\left(\frac{1}{(1+\delta_a)^2} + (1+\delta_a)^2 + 4\right) \approx \frac{\hbar^2\pi^2}{2md^2}(6 + 4\delta_a^2)$$

$$E_{121} = \frac{\hbar^2\pi^2}{2md^2}\left(\frac{1}{(1+\delta_a)^2} + 4(1+\delta_a)^2 + 1\right) \approx \frac{\hbar^2\pi^2}{2md^2}(6 + 6\delta_a)$$

$$E_{121} = \frac{\hbar^2\pi^2}{2md^2}\left(\frac{4}{(1+\delta_a)^2} + (1+\delta_a)^2 + 1\right) \approx \frac{\hbar^2\pi^2}{2md^2}(6 - 6\delta_a)$$

c)

The degeneracy of the first excited state will be lifted. It will split into 3 different energy levels.

d)

Cubic box:

$$E_{112} = \frac{\hbar^2\pi^2}{2md^2}6$$

$$E_{121} = \frac{\hbar^2\pi^2}{2md^2}6$$

$$E_{211} = \frac{\hbar^2\pi^2}{2md^2}6$$

The same energy for all 3 states

e)

Single peak will split into 3 closely spaced peaks. The position of the middle peak will remain the same as the position of the single peak in case of the cubic box.